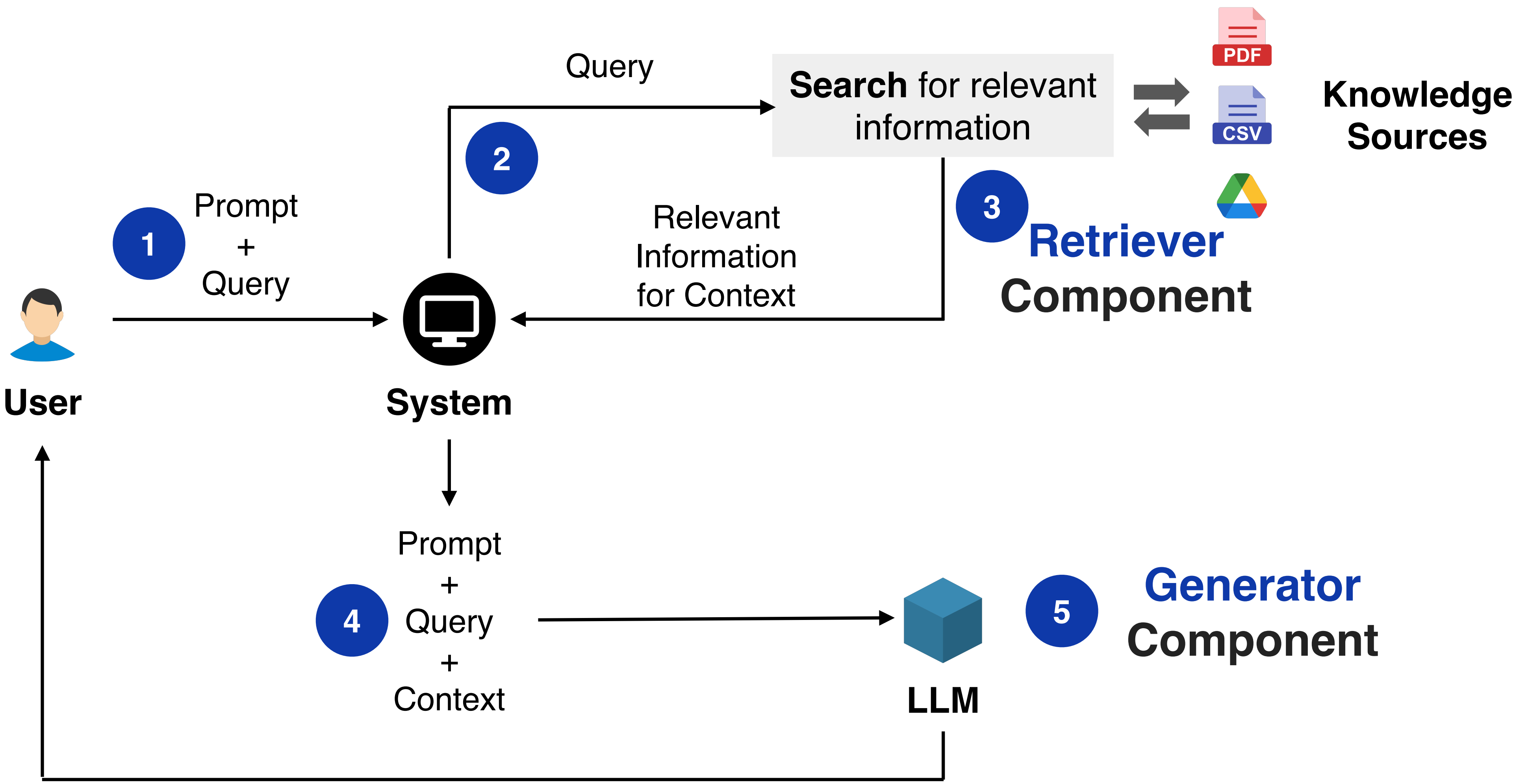
Agentic RAG

From Retrieval to Reasoning

Recap: Vanilla RAG



Limitations: Vanilla RAG

- Linear Pipeline

  It performs a single search based on the initial query and generates a response based solely on those results.

- The "Lost in the Middle" Phenomenon

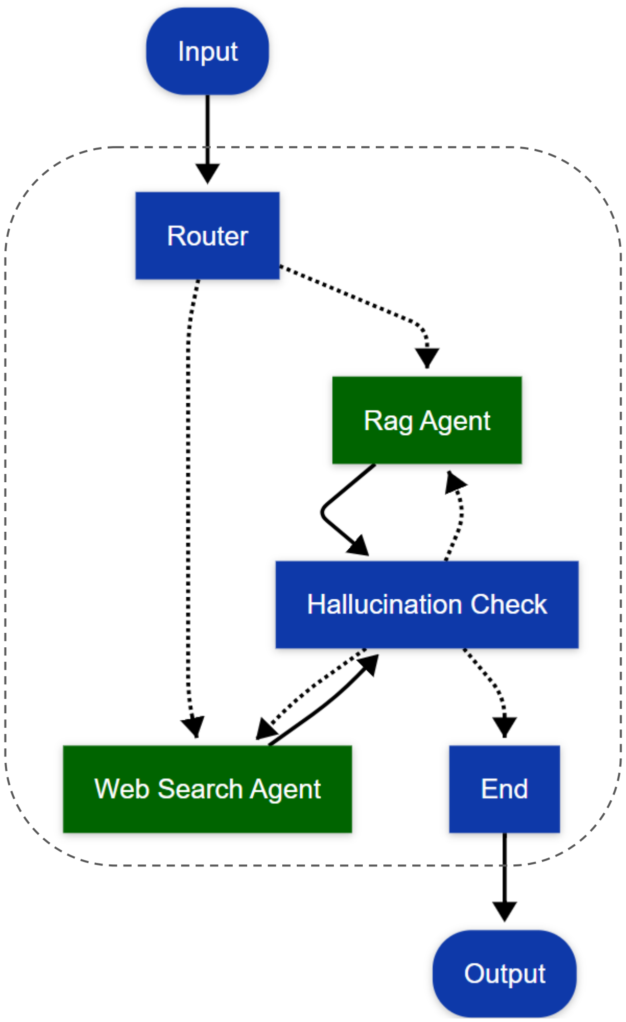
  It tends to prioritize information at the very beginning and very end of a prompt. Important when prompts are contexts are long.

- Knowledge Availability Assumption

  It assumes the information needed exists within a single set of retrieved "chunks."

Agentic RAG

Specialized agents handle distinct phases. Workflow contains nodes other than the agents (e.g., router, judge, rewriter, querry refiner, hallucination checker, policy/safety filter).



- Intelligent Routing: A router agent first directs the query to the most suitable information retrieval agent.
- Specialized Retrieval: A Rag Agent (for internal knowledge) or a Web Search Agent (for broader, real-time info) gathers and generates an initial response.
- Rigorous Verification: A dedicated hallucination check function critically evaluates the answer for factual accuracy and grounding (note: this node is not an agent).
- Iterative Refinement & Output: The system can loop back for re-retrieval or refinement if inaccuracies are detected, or deliver the verified final answer closing the loop.

Vanilla RAG

- “Should AI be regulated like pharmaceuticals?”
- User >> Retriever (Vector DB) >> Top-k documents >> LLM (Answer generation) >> User
- One-shot retrieval
- No counterargument exploration
- No iterative refinement
- No explicit grounding verification

Agentic RAG (Single-Agent + Verifier Loop)

- Introduce a control logic
- User >> Router >> RAG Agent >> Grounding Verifier (LLM Judge) >>
    - If PASS >> User
    - If FAIL >> RAG Agent (retry)
- RAG Agent can
    - Reformulate query
    - Retrieve more documents
    - Regenerate answer
- Agentic RAG introduces iteration + validation.

Multi-Agent Agentic RAG

- Two Grounded Agents: Pro and Con
  - Each must: Retrieve evidence, Cite documents and Defend claims
  - Graph architecture, with other agents (judge, fact checker etc)
  - Retrieval + structured adversarial reasoning
- Each debate agent is a mini Agentic RAG system.
- For each round, each Agent:
  - Reformulate argument
  - Call Retriever tool
  - Select supporting documents
  - Generate cited argument
- RAG powered Debate System